

ANJALI DESAI

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SUMMARY

High school student with published research in Artificial Intelligence and computer science. Published researcher in International Journal of Scientific Research (air quality forecasting using GRU models) with ongoing work in breast cancer classification using multimodal deep learning. Strong academic background (97% in 9th, 96.4% in 10th) with international recognition: AMO Global Rank 29, UCO national topper. Tech Lead for Shakti Society, Vice-Director of Digital Innovation, and volunteer educator with CRY foundation.

EDUCATION

Physics, Chemistry, Mathematics, Computer Science (CBSE)

Graduating 2027

National Hill View Public School, Bangalore, India

9th: 97%, 10th: 96.4% (97% with AI elective)

Relevant coursework: Mathematics, Physics, Chemistry, Artificial Intelligence, Computer Science, Programming (Java, C++, Python)

TECHNICAL SKILLS

Programming Languages: Python, Java, C++, HTML/CSS, JavaScript

AI/ML Frameworks & Tools: PyTorch, TensorFlow, scikit-learn, Keras, Pandas, NumPy, Matplotlib, Seaborn

Development Tools: GitHub, Google Colab, Jupyter Notebook, Streamlit, VS Code

Certifications: Brown University AI/ML & Data Science (98%), Java Specialization (Codio), Python (Google), C++ Specialization (Codio)

Research Areas: Deep Learning, Computer Vision, Medical AI, Time Series Forecasting, Multimodal Learning

RESEARCH & PUBLICATIONS

AI-Based Air Quality Index (AQI) Forecasting in Bengaluru Using GRU Models

Published 2025

Published in International Journal of Scientific Research (IJSR)

- Conducted independent research on deep learning applications for temporal air pollution forecasting in Bengaluru.
- Developed and evaluated GRU-based neural networks using historical air quality datasets; demonstrated significant improvement over traditional forecasting models.
- Published in International Journal of Scientific Research (IJSR), June 2025; also featured on OneYoungIndia publication platform.
- Available at: [https://www.worldwidejournals.com/international-journal-of-scientific-research-\(IJSR\)/recent_issues_pdf/2025/June/air-quality-index-forecasting-in-bangalore-using-a-gru-deep-learning-June_2025_4561939467_6530047.pdf](https://www.worldwidejournals.com/international-journal-of-scientific-research-(IJSR)/recent_issues_pdf/2025/June/air-quality-index-forecasting-in-bangalore-using-a-gru-deep-learning-June_2025_4561939467_6530047.pdf)

Multimodal Deep Learning for Breast Cancer Subtype Classification

July 2024 – October 2024

Research paper in progress; independent research with AGSD mentorship

- Developed multimodal deep learning framework integrating histopathology images and biomarker data to classify four breast cancer subtypes (IDC, ILC, TNBC, MBC).
- Led dataset curation (600 samples) and model architecture design, testing ResNet18, ResNet50, and DINO Vision Transformer configurations.
- Achieved 87% relative improvement in macro-F1 score (0.678 vs 0.363 baseline) by integrating clinically relevant biomarkers with imaging data.
- Conducted data preprocessing, model training (20 epochs), performance evaluation, and developed Streamlit deployment interface.
- Worked approximately 7 hours/week as part of 8-person research team; manuscript in preparation for submission.

ACADEMIC AND RESEARCH EXPERIENCE

Brown University Pre-College AI, ML, and Data Science Program, Online: Team Leader & Capstone Project Lead Summer 2024

- Completed intensive AI, Machine Learning, and Data Science program with 98% final grade.
- Led team of four students in developing end-to-end stock market prediction model using machine learning for program capstone.
- Applied supervised learning, time series analysis, and model evaluation techniques; achieved over 86% directional accuracy in backtesting.
- Demonstrated initiative in project planning, peer collaboration, and technical presentation skills.

AGSD (ALLEN Global Studies Division) AI Bootcamp, Online: Participant August 2024 – November 2024

- Completed structured 12-week bootcamp focused on foundational and advanced concepts in AI, ML, and Python programming.
- Participated in weekly coding challenges and AI mini-projects; gained practical exposure to model development, data preprocessing, and evaluation.
- Received mentorship on breast cancer classification research project from advisor with biomarker expertise.

LEADERSHIP AND TECHNOLOGY INITIATIVES

Vice-Director of Digital Innovation 2025 – Present

Elected student leader for technology and AI initiatives at National Hill View Public School

- Leading school-wide technology initiatives to promote AI integration in classroom learning and peer-led education.
- Spearheading launch of AI-focused club with workshops, mini-projects, and competitions to build student technical skills.
- Mentoring junior students in Python, HTML, and AI basics; coordinating with external mentors to provide real-world AI exposure.

Tech Lead – Shakti Society 2024 – Present

Student-led women’s empowerment and safety initiative

- Developed and maintain full-stack website for women’s empowerment organization featuring event management, resource library, and social media integration.
- Manage technical infrastructure and expand platform features to support community outreach and engagement.
- Lead technology strategy for organization’s digital presence and online programs.

Tech and Event Management Team – AETHRA 2024 – Present

Technology organization focused on hackathons and digital literacy

- Organize and coordinate hackathons and digital literacy workshops for student community.
- Manage technical logistics and event planning for technology education initiatives.
- Contribute to expanding digital literacy access through structured workshops and events.

ACHIEVEMENTS AND RECOGNITION

American Mathematics Olympiad (AMO), Global: Top Performer 2024

- Global Rank: 29, Country Rank: 6, Global Finals Rank: 14
- Competed in challenging problem-solving across algebra, geometry, number theory, and combinatorics.

Unified Cyber Olympiad (UCO), India: National Topper Multiple Years

- Achieved 1st and 2nd place national rankings across multiple years in computer science and cybersecurity olympiad.
- Demonstrated excellence in programming fundamentals, computational thinking, and digital literacy.

Global Essay and Music Awards (GEMA), National: Winner

2023

- Won national short story competition; winning story published as prize.
- Recognized for creative writing excellence in competitive literary arts.

Standardized Testing, :

2024

- SAT Score: 1550 (English: 770, Math: 780)

COMMUNITY SERVICE AND VOLUNTEERING

Child Rights and You (CRY) Foundation, Bangalore, India: Digital Literacy Volunteer

2023 – Present

- Contributed over 60 hours per semester teaching cybersecurity awareness, digital literacy, and online safety to underprivileged students.
- Designed and delivered structured educational modules on digital marketing and 21st-century employable skills.
- Led volunteer teams during community events including Walk to EmpowHER walkathon and World Music Day celebrations.

CREATIVE AND ARTISTIC PURSUITS

Creative Writing – Blog and Kindle eBook

2023 – Present

GEMA award winner; published author with consistent readership

- Write and publish original fiction twice monthly exploring historical fantasy, psychological suspense, and urban supernatural genres.
- Built readership through blog (<https://writerjourneyy.blogspot.com/>) and formatted selected works into Kindle e-book.
- Won GEMA national short story competition and multiple inter-school creative writing awards.

Music Performance – Dhuniscian

Ongoing

Trinity Grades 1–3 (Outstanding); Grade 4 ongoing

- Three years of training in guitar and keyboard; proficient in classical and contemporary styles.
- Active performer with Dhuniscian music group at school events, band performances, and cultural programs.
- Created YouTube channel (<https://www.youtube.com/@anjalidesai>) showcasing music compositions and covers; actively building online portfolio.